

Stage-3 – Reponses LLM (A vs C)

Texte = debut de la reponse FINALE de l'agent (result_head capture par run_live),
cape a ~2500 caracteres par le harness ; les appels d'outils intermediaires ne sont
pas inclus. A = Revit-direct pur (no-KG). C = KG v2 (projection + read-API).

Une page par scenario ; colonne gauche = reponse A, colonne droite = reponse C.
Chiffres et verdicts : voir le document principal (stage3-all-scenarios.pdf).

Scenarios (ordre des pages) :

Principaux	Audit-XL (70)	70_audit-xl.txt
	Fanout-XL (75)	75_fanout-xl.txt
	M-Long (80)	80_m-long.txt
	Drift-Long (85)	85_drift-long.txt
	Resume (90)	90_resume.txt
	Audit-Trail (95)	95_audit-trail.txt
Neutres	P1	10_P1.txt
	S4	40_S4.txt
	Q	60_Q.txt
	M	70_M.txt
Graphe	Impact-XL	10_impact-xl.txt
	TypeImpact-XL	20_typeimpact-xl.txt
	Provenance	30_provenance.txt

Audit-XL (70) – Principaux (DESIGN kg-v2 §8.3)

Reponse A (Revit-direct)

Audit complete – walked the full window → host wall → level chain in one read-only pass over the live model.

Audit-XL results

****1. Total windows checked: 352****

****2. Violating windows (25)**** – every one fails on ****sill**** only (sill_height 0.2 m < 0.9 m). No window is hosted on a below-grade wall, so there are no `level` or `both` violations (the only sub-zero level, *Niveau -1* @ -3 m, hosts no windows).

Window id	Host wall id	Level (elevation)	sill_height	Violates
256089	255080	Niveau 0 (0 m)	0.2 m	sill
256342	255449	Niveau 0 (0 m)	0.2 m	sill
257355	257345	Niveau 1 (3 m)	0.2 m	sill
257373	257347	Niveau 1 (3 m)	0.2 m	sill
257377	257347	Niveau 1 (3 m)	0.2 m	sill
257426	257411	Niveau 2 (6 m)	0.2 m	sill
257441	257413	Niveau 2 (6 m)	0.2 m	sill
257462	257417	Niveau 2 (6 m)	0.2 m	sill
257463	257417	Niveau 2 (6 m)	0.2 m	sill
257495	257477	Niveau 3 (9 m)	0.2 m	sill
257508	257479	Niveau 3 (9 m)	0.2 m	sill
257527	257483	Niveau 3 (9 m)	0.2 m	sill
257559	257543	Niveau 4 (12 m)	0.2 m	sill
257571	257545	Niveau 4 (12 m)	0.2 m	sill
257587	257547	Niveau 4 (12 m)	0.2 m	sill
257636	257611	Niveau 5 (15 m)	0.2 m	sill
257642	257611	Niveau 5 (15 m)	0.2 m	sill
257684	257677	Niveau 6 (18 m)	0.2 m	sill
257691	257677	Niveau 6 (18 m)	0.2 m	sill
257702	257679	Niveau 6 (18 m)	0.2 m	sill
257703	257679	Niveau 6 (18 m)	0.2 m	sill
257704	257679	Niveau 6 (18 m)	0.2 m	sill
257720	257681	Niveau 6 (18 m)	0.2 m	sill
257728	257683	Niveau 6 (18 m)	0.2 m	sill
257788	257747	Niveau 7 (21 m)	0.2 m	sill

****3. Summary: 25/352 windows violate the rule.****

Nothing was modified. (Note: elevations are reported in meters, converted from Revit's internal feet; the French-locale model returns `0,2` which is 0.2 m.)

Reponse C (KG v2)

Confirmed: total windows = ****352**** (matches the join exactly, so no window was dropped from the chain), and Niveau -1 hosts ****zero**** windows – so the level clause catches nothing. All 25 violators are sill-only.

Audit-XL – Compliance Report

****1. Total windows checked:** 352**

****2. Violating windows** (25)** – all violate the ****sill**** rule (sill_height 0.2 m < 0.9 m). The only level with elevation < 0 m (Niveau -1, -3 m) hosts no windows, so no level/both violations exist.

Window	Host wall	Level (elevation)	Violation
window_010	wall_002	Niveau 0 (0 m)	sill
window_044	wall_008	Niveau 0 (0 m)	sill
window_048	wall_062	Niveau 1 (3 m)	sill
window_066	wall_064	Niveau 1 (3 m)	sill
window_070	wall_064	Niveau 1 (3 m)	sill
window_097	wall_072	Niveau 2 (6 m)	sill
window_112	wall_074	Niveau 2 (6 m)	sill
window_131	wall_078	Niveau 2 (6 m)	sill
window_132	wall_078	Niveau 2 (6 m)	sill
window_144	wall_082	Niveau 3 (9 m)	sill
window_157	wall_084	Niveau 3 (9 m)	sill
window_174	wall_088	Niveau 3 (9 m)	sill
window_186	wall_092	Niveau 4 (12 m)	sill
window_198	wall_094	Niveau 4 (12 m)	sill
window_212	wall_096	Niveau 4 (12 m)	sill
window_241	wall_104	Niveau 5 (15 m)	sill
window_247	wall_104	Niveau 5 (15 m)	sill
window_265	wall_112	Niveau 6 (18 m)	sill
window_272	wall_112	Niveau 6 (18 m)	sill
window_283	wall_114	Niveau 6 (18 m)	sill
window_284	wall_114	Niveau 6 (18 m)	sill
window_285	wall_114	Niveau 6 (18 m)	sill
window_299	wall_116	Niveau 6 (18 m)	sill
window_307	wall_118	Niveau 6 (18 m)	sill
window_345	wall_126	Niveau 7 (21 m)	sill

****3. Summary:** 25/352 windows violate the rule.**

Nothing was modified. Method: one `kg\_v2\_query` join walked window → host wall → level for all 352 windows; sill values are only ever 0.9 or 0.2, so the 25 windows with sill_height 0.2 are the sill violators, and a traversal from the only negative-elevation level (Niveau -1) confirmed it hosts no windows.

Fanout-XL (75) – Principaux (DESIGN kg-v2 §8.3)

Reponse A (Revit-direct)

All data retrieved from the live model in one read-only query (84 walls, 352 windows with host wall, level, type, height, centerline length).
Computed answers:

Q1. 84 walls (alive, `WhereElementIsNotElementType`).

Q2. No single winner – an 8-way tie at 10 walls each: Niveau 0, 1, 2, 3, 4, 5, 6, 7 all host 10 walls (Niveau -1 hosts only 4). If a single answer is forced, any of those eight levels with 10 walls.

Q3. Two distinct WallTypes in use:

WallType	Wall instances
--- ---	
Générique - Ext. 200 mm	36
cloison	48

Q4. Mean wall height = 2.85 m (32 walls @ 3.00 m + 52 walls @ 2.75 m → 239/84 = 2.845).

Q5. 0 orphan windows – all 352 windows are hosted by a wall (`fi.Host is Wall` true for every one).

Q6. 52 walls have no window. Only 32 distinct walls host windows (4 per level × 8 levels); 84 - 32 = 52.

Q7. Average windows per wall, by level:

Level	windows/wall
--- ---	
Niveau 0	4.40 (44 win / 10 walls)
Niveau 1	4.40
Niveau 2	4.40
Niveau 3	4.40
Niveau 4	4.40
Niveau 5	4.40
Niveau 6	4.40
Niveau 7	4.40
Niveau -1	0.00 (0 win / 4 walls)

Q8. None. Every wall is either 3.00 m or 2.75 m; no wall is strictly > 3.0 m.

Q9. No. Wall lengths are only 14.0, 16.0, 7.8, and 4.1 m; the shortest is 4.1 m, so none is < 1.0 m.

Q10. Sum of (length × height) = 2744.6 m².
(Generic walls: 7 levels×60 m×3.00 + 1 level×60×2.75 + Niveau -1 60×3.00 = 1605; cloison: 8 levels × 51.8 m × 2.75 = 1139.6.)

10/10 answered.

Reponse C (KG v2)

All figures confirmed by server-side aggregation. Here are the ten answers, all derived from the KG v2 read tools (project `500\_objs`, turn 52, 84 alive walls / 352 windows / 10 levels).

Q1 – Total alive walls: 84

Q2 – Level with most walls: It's an 8-way tie at 10 walls each: Niveau 0, 1, 2, 3, 4, 5, 6 and 7 all host 10 walls. (Niveau -1 has 4; Toit has 0.) No single maximum.

Q3 – Distinct WallTypes in use (instance count):

WallType	Walls
--- ---	
`cloison` (revit_257865)	48
Générique - Ext. 200 mm` (revit_29331)	36

Q4 – Mean wall height: 2.85 m (2.84524..., = (32×3.0 + 52×2.75)/84).

Q5 – Orphan windows (no host wall): 0 – all 352 windows resolve to a host wall (32 distinct host walls, no null-host group).

Q6 – Walls with no window hosted: 52 (84 total - 32 walls that host windows).

Q7 – Average windows per wall, by level (windows carry no level attr, so derived via host-wall → level; each populated level has 2 walls×15 + 2 walls×7 = 44 windows over 10 walls):

Level	Avg windows/wall
--- ---	
Niveau 0-7 (each)	4.40
Niveau -1	0.00 (4 walls, 0 windows)
Toit	n/a (0 walls)

Q8 – Walls with height strictly > 3.0 m: None. The maximum height in the model is exactly 3.0 m.

Q9 – Walls with length strictly < 1.0 m: None – the minimum wall length is 4.1 m.

Q10 – Σ(length × height) over all alive walls: 2744.6 m² (= 3.0×480 + 2.75×474.4 = 1440 + 1304.6).

10/10 answered.

M-Long (80) – Principaux (DESIGN kg-v2 §8.3)

Reponse A (Revit-direct)

All 30 edits applied. Each returned `succeeded:2, failed:0, rolled\_back:false`, so every height write committed to the live Revit model.

Final differential – all 30 edits

Differential reconstructed from my 30 confirmed write-transactions (the live model has no history/diff primitive, and no project-memory layer is in scope). Wall # = N-th wall by ascending ElementId.

#	ElementId	height (m)	#	ElementId	height (m)
1	255040	2.80	16	257349	2.60
2	255080	2.81	17	257350	2.61
3	255124	2.82	18	257351	2.62
4	255179	2.83	19	257382	2.63
5	255289	2.84	20	257383	2.64
6	255352	2.85	21	257410	2.65
7	255396	2.86	22	257411	2.66
8	255449	2.87	23	257412	2.67
9	256196	2.88	24	257413	2.68
10	256298	2.89	25	257414	2.69
11	257344	3.10	26	257415	3.20
12	257345	3.12	27	257416	3.21
13	257346	3.14	28	257417	3.22
14	257347	3.16	29	257448	3.23
15	257348	3.18	30	257449	3.24

Note: each edit also set the wall's Top Constraint to **Unconnected** – the raw `WALL\_USER\_HEIGHT\_PARAM` was read-only while the top was level-constrained, so detaching it was required to set an explicit height.

30/30 edits persisted.

Reponse C (KG v2)

All 30 target walls now carry their exact target heights in the live KG projection – including the 7 first-batch walls I'd flagged at the checkpoint (wall_005-010, 065), which projected correctly. The diff noise was just Revit's regeneration cascade, not a persistence failure.

Final differential summary – all 30 edits

KG-verified projected `height` (m), edits 1-30, walls in `llm\_id` order:

#	wall	h	#	wall	h	#	wall	h
1	wall_001	2.80	11	wall_061	3.10	21	wall_071	2.65
2	wall_002	2.81	12	wall_062	3.12	22	wall_072	2.66
3	wall_003	2.82	13	wall_063	3.14	23	wall_073	2.67
4	wall_004	2.83	14	wall_064	3.16	24	wall_074	2.68
5	wall_005	2.84	15	wall_065	3.18	25	wall_075	2.69
6	wall_006	2.85	16	wall_066	2.60	26	wall_076	3.20
7	wall_007	2.86	17	wall_067	2.61	27	wall_077	3.21
8	wall_008	2.87	18	wall_068	2.62	28	wall_078	3.22
9	wall_009	2.88	19	wall_069	2.63	29	wall_079	3.23
10	wall_010	2.89	20	wall_070	2.64	30	wall_080	3.24

Notes (honest):

- "wall N" was mapped to the **first 30 walls in `llm\_id` order** (wall_001-010 then wall_061-080); the fixture has no live wall_011-060, so positional N≠wall_0NN.
- 12 perimeter walls (1-4, 11-14, 21-24) were top-constrained "Up to level", so I flipped their Top Constraint to **Unconnected** before setting height – otherwise `WALL\_USER\_HEIGHT\_PARAM` is read-only.
- I used **2 batched transactions** (split at the checkpoint) rather than 30 separate ones, per the "fewest tool calls" directive – KG projection is automatic and per-element, so all 30 persisted; they coalesced into KG turns 53-54.

30/30 edits persisted

Drift-Long (85) – Principaux (DESIGN kg-v2 §8.3)

Reponse A (Revit-direct) [tronquee a 2500c]

Both edits committed (2/2 succeeded). Here is the full audit.

Drift-Long audit – Revit-direct profile (no project-memory layer)

****Up-front constraint that governs the whole audit:**** this profile has no `store\_\*/kg\_\*/persistence` layer. My only source of truth is the live Revit document, re-queried each turn. That single fact determines Steps 2, 3 and 5 – see below.

Step 1 – Baseline (live query)

Element	Live count	Notes
Walls	****≥ 50**** (`ai\_element\_filter` returned exactly 50 = the query page cap)	Clean structure of 10 walls × levels N0-N4; more may exist on upper levels beyond the page
Windows	****≥ 50**** (also returned at the 50 cap)	type "0.60 × 0.95 m", on N0/N1 in the returned page
Levels	****10**** (exact, under cap)	Niveau -1, 0, 1, 2, 3, 4, 5, 6, 7, Toit

There is ****no turn-id / snapshot-id**** to report: snapshot identifiers are a property of a memory/versioning layer, and none exists here. Each query is a fresh live read with no notion of "as of turn N."

Step 2 – Detect drift

****I cannot detect drift, by construction – and this is the honest, expected answer for this profile.**** Drift means a divergence between **remembered/tracked state** and **live state**. Here there is no tracked state to diverge from: I hold no prior snapshot, no per-element baseline, no history. Every read returns whatever Revit currently contains, so live-vs-tracked comparison is undefined.

Concretely, the three drift classes the scenario asks about are

****undetectable**** with live-only tooling:

- ****Deleted out-of-band**** (present before, gone now): a deleted element simply doesn't appear in the query. With nothing to compare against, I cannot know it was ever there.
- ****Added out-of-band****: a new element is indistinguishable from one that "should" be there – I have no expected set.
- ****Attribute change**** (e.g. a sill lowered, a wall moved/heightened): I see only the current value, never the "before," so I cannot say it changed.

The element heights in the live model do vary (e.g. walls stepping 2800→2810→2820 mm, upper-level partitions at 2750 mm), but ****I cannot attribute any of this to drift**** – it's equally consistent with the model's authored design. Calling it "drift" would be fabrication.

Step 3 – Reconcile vs flag

Nothing to act on: zero drifts were **detectable** in Step 2. There is no tracked state to reconcile, and reconciliation isn't a meaningful operation when the live model is itself the only source of truth.

Reponse C (KG v2)

Audit complete. Here is the six-step report.

Drift-Long audit – project `500\_objs` (project_id `42d0cc796791a952`)

****Step 1 – Baseline**** (turn 55 at start)

- Walls: ****84**** · Windows: ****352**** · Levels: ****10****
- KG v2 totals: 521 nodes / 1254 edges · pending deltas: 0 · last action: `turn 54: modify wall\_087`

****Step 2 – Drift detection****

`kg\_v2\_detect\_drift` checked all ****471**** projected instance nodes against the live Revit document. ****0 divergences**** – no `missing\_in\_kg`, no `orphan\_kg\_node`, no `tombstoned\_but\_live`, no `attrs\_diverged`. No out-of-band edit was applied to the model since the last projection. Note: drift detection here is **constructive** – it diffs the KG projection directly against the live doc element-by-element, so it would surface any UI edit regardless of whether the DocumentChanged watcher observed it live.

****Step 3 – Reconcile vs flag****

Nothing to act on – zero drifts. No reconciliation or flagging performed.

****Step 4 – Batch edit**** (applied via `batch\_set\_parameters`, one atomic transaction)

- `wall\_001` (Revit id 255040): height 2.8 → ****2.9 m**** (+0.1) 
- `wall\_002` (Revit id 255080): height 2.81 → ****2.71 m**** (-0.1) 
- 2/2 succeeded, 0 failed. KG projection auto-tracked both (each node now carries `modified\_at\_turn: ..., 55` and the new heights), confirming the DocumentChanged hook stayed in sync.

****Step 5 – Drift detection (re-run)****

`kg\_v2\_detect\_drift`: ****471 checked, 0 drift****. The tool-authored edits left projection and live model consistent – no self-inflicted divergence.

****Step 6 – Final summary****

- Total drifts surfaced: ****0**** (0 in Step 2 + 0 in Step 5)
- Reconciled: 0 · Flagged: 0
- Verdict: ****project consistent.****

Resume (90) – Principaux (DESIGN kg-v2 §8.3)

Reponse A (Revit-direct)

Verified: walls ****86**** (was 84), and the ****Toit**** level (highest elevation) went from 1 → 3 elements – both new walls landed there and persisted.

****Step 1 – Active project****
- ****Project:**** `500\_objs` (queried via `analyze\_model\_statistics`).
- ****Totals:**** 4,033 elements / 515 types across 41 categories before my edit; 84 walls, 10 levels.
- ****No turn/snapshot index exists.**** The Revit-direct tools expose only the **current** document state – there is no project-memory layer, no snapshot counter, no session id. Counts above come from `analyze\_model\_statistics` + `ai\_element\_filter`.

****Step 2 – Prior session's diff: cannot be recovered (stated limitation)****
The live Revit model carries ****no cross-session history****. By construction the only things I can read are the elements as they exist **now** – there is no persisted change log, no per-element creation/modification timestamp, no "old value" record, and no deletion tombstones. A saved-and-reopened `.rvt` discards the undo stack, so the "≈20 added / handful modified" deltas from the previous session are ****not retrievable**** through `ai\_element\_filter`/`get\_\*`. I will not fabricate a diff.

The closest **inference** the live model permits is ElementId ordering (Revit ElementIds are monotonic, so higher ≈ created later): the highest-Id walls (257827–257830 on a "Niveau -1" added under a distinct document GUID prefix `c2209253...`, vs. the bulk `991bbc90...`) are consistent with a recent editing episode. But that is a heuristic, not an authoritative diff – I cannot confirm which elements were modified, nor their prior values, from the live model alone.

****Step 3 – Continued the work****
- Highest-elevation level = ****Toit**** (≈23,750 mm; above Niveau 7 at 21,000 mm).
- "Most recently created wall" taken as the max-ElementId wall = ****257830****, type ****Générique - Ext. 200 mm**** (TypeId 29331) – again an ElementId-order inference, the only basis the live model offers.
- Created ****2 walls**** on Toit, each 3,000 mm long × 2,700 mm high, type 29331 (`create\_line\_based\_element`) → IDs ****257991, 257992****. Verified live: walls 84→86, Toit 1→3 elements.

session resumed: 0 prior changes recovered, 2 new walls added

(0 because the live model exposes no cross-session history – the prior diff is unrecoverable by construction, not empty.)

Reponse C (KG v2) [tronquee a 2500c]

Both walls are confirmed in the KG projection – type `revit\_29331` ("Générique - Ext. 200 mm"), level_ref `revit\_257800` (Toit), length 3 m, height 2.7 m. Node count rose 521→523 (+2 walls), edges 1254→1258 (+4: each wall's `of\_type` + `at\_level`), `PendingDeltaCount: 0` (persisted to ES). Here is the full report.

Step 1 – Situate

Tool used: `kg\_v2\_session\_info`.

Field	Value
Project id	`42d0cc796791a952`
Document	`500_objs`
Turn at entry	**56**
Nodes / edges	**521 / 1254**
Pending deltas	0 (clean, fully flushed)
Last action	"turn 55: modify wall_009"

Step 2 – Recover the prior diff (and an honest limitation)

Tools used: `kg\_v2\_diff\_since(since\_turn=0)`, then targeted reads/greps of the action log, plus `kg\_v2\_get\_by\_revit\_id`.

****Cross-session history *is* reachable.**** Unlike the old v1 KG, the v2 append-only ES journal persisted intact (827 KB), so I can read every action back to turn 1. So I'm **not** in the "can't reach back" branch – I can reach back.

****But the log carries no session boundary, and what it actually contains contradicts the briefing.**** The full beginning→now diff is one continuous turn stream (1→55) recording the **entire** 500-object model build:

- ****485 `create`**** and ****50 `delete`**** operations – ****all confined to turns ≤52**** (the bulk seed; the last create is `wall\_133`, turn 52).
- ****Turns 53, 54, 55 are exclusively `modify`**** – ****zero creates, zero deletes after turn 52.****

This matters for your briefing's claim of "≈twenty new elements added." ****I cannot independently confirm that**** – there is no create burst at the tail; all creations are indistinguishable from the bulk seed. Per the instruction not to assume what the tools can't confirm, ****I will not fabricate a 20-element creation diff.****

What **is** independently confirmed as the most-recent ("prior session") activity is ****modifications only**** – wall-height edits, e.g.:
- `wall\_001`: 3.0 → 2.8 m (turn 53), then 2.8 → ****2.9 m**** (turn 55)
- `wall\_002`: 3.0 → 2.81 m (turn 53), then 2.81 → ****2.71 m**** (turn 55)
- `wall\_003`: 3.0 → 2.82 m (turn 53), and a similar run across other walls

Note: many turn-55 entries (all 30 windows, several walls) are ****no-op re-projection echoes**** where `before == after`, not real changes – a known artifact of the DocumentChanged bulk re-projection.

Audit-Trail (95) – Principaux (DESIGN kg-v2 §8.3)

Reponse A (Revit-direct)

All ten replacements are done. Now the audit-trail requirement.

Replacements (done via Revit-direct tools)

Cycle	Old wall id (deleted)	New wall id (created)	Type swap
1	255040	257995	Générique 200 mm → cloison
2	255080	257996	Générique 200 mm → cloison
3	255124	257997	Générique 200 mm → cloison
4	255179	257998	Générique 200 mm → cloison
5	255289	257999	cloison → Générique 200 mm
6	255352	258000	cloison → Générique 200 mm
7	255396	258001	cloison → Générique 200 mm
8	255449	258002	cloison → Générique 200 mm
9	256196	258003	cloison → Générique 200 mm
10	256298	258004	cloison → Générique 200 mm

Each new wall was created at the same centerline, level (Niveau 0), and height as the wall it replaced, but with a different wall type. The old ids are gone (the delete cascaded 54 elements – the 10 walls plus their hosted children).

Audit trail – cannot be recorded (this is case #3)

I must state this explicitly rather than invent a relation:

- **No project-memory layer is available to me.** In this profile I author/query the model only through Revit-direct tools (``create_*/`modify_*/`delete_*`, ``ai_element_filter`/`get_*`). I am not permitted to use ``store_*_data``, ``kg_*`, or any persistence/memory tool, so there is nowhere to record a "replaces" relation, a reason string, or a cycle timestamp.
- **The live Revit model has no concept of a semantic "replaces" relation** between a deleted element and a new one. Once a wall is deleted, its ``ElementId`` is gone and is no longer queryable; Revit retains no link from the new wall back to the old one, no per-element creation timestamp/turn, and no free-text "reason" field for an inter-element relation. The mapping in the table above exists only in this transcript – it is **not** queryable from the model.

Consequently, requirement (1) "enumerate the recorded 'replaces' relations" and the second half of requirement (2) "the audit trail relating them to their replacements is still queryable" **cannot** be satisfied** with the tools I have. I can confirm the first half of (2): the old walls (255040, 255080, 255124, 255179, 255289, 255352, 255396, 255449, 256196, 256298) are no longer alive – they were deleted – but no audit trail relating them to the new walls is queryable.

10/10 replacements done, 0/10 with audit trail recorded.

Reponse C (KG v2) [tronquee a 2500c]

All ten cycles are complete and verified. KG v2 **does** represent a semantic supersession relation (the ``replaced_by`` annotation kind), so no work was blocked.

Audit trail – ``replaced_by`` relations

Each old wall was deleted and re-created at the identical line/level with a different wall type, then linked old→new via a ``replaced_by`` annotation. All edges recorded at **turn 69**; geometry confirmed identical between each old wall and its replacement.

Cycle	Old wall (id / revit_id)	New wall (id / revit_id)	Type swap	Reason	Turn
1	wall_001 / 255040	wall_137 / 257968	29331→29332	(Ext.200→Ext.150) "type swap, cycle 1"	69
2	wall_002 / 255080	wall_138 / 257969	29331→29332	"type swap, cycle 2"	69
3	wall_003 / 255124	wall_139 / 257970	29331→29332	"type swap, cycle 3"	69
4	wall_004 / 255179	wall_140 / 257971	29331→29332	"type swap, cycle 4"	69
5	wall_005 / 255289	wall_141 / 257972	257865→29644	(cloison→Int.70) "type swap, cycle 5"	69
6	wall_006 / 255352	wall_142 / 257973	257865→29644	"type swap, cycle 6"	69
7	wall_007 / 255396	wall_143 / 257974	257865→29644	"type swap, cycle 7"	69
8	wall_008 / 255449	wall_144 / 257975	257865→29644	"type swap, cycle 8"	69
9	wall_009 / 256196	wall_145 / 257976	257865→29644	"type swap, cycle 9"	69
10	wall_010 / 256298	wall_146 / 257977	257865→29644	"type swap, cycle 10"	69

The recorded "timestamp" is the KG turn counter (turn 69 for all annotations); the underlying delete happened at turn 58 and creates at turns 59–68, all retrievable from the action log via ``kg_v2_diff_since``.

Confirmations

- Old walls are dead.** wall_001-wall_010 all carry ``deleted_at_turn: 58`` and are hidden from a normal query – they no longer exist as live Revit elements (their delete also cascaded to 44 hosted windows/openings, 54 elements total).
- Audit trail is still queryable.** With ``include_soft_deleted: true`` the ten tombstones remain in the graph, and ``kg_v2_diff_since(68)`` returns all ten ``replaced_by`` edges with their cycle reasons – so the supersession relation linking each dead wall to its live replacement survives the deletion and is independently retrievable from the KG, not just from this session's memory.
- Not applicable – the relation is fully represented by the ``replaced_by`` annotation kind; nothing was invented.

**10/10 replacements done, 1

P1 – Neutres (§8.4-iv)

Reponse A (Revit-direct)

Done. All six walls now verified on N0, base at z=0, with wall #3 at 2900 mm and the rest at 2700 mm.

Report (from live model query)

Counts – whole project (model `500\_objs` is pre-populated, 4006 elements):
- **Levels:** 12 total. I added 2 (N0 @ 0 mm, N1 @ 3000 mm); the other 10 pre-existed (Niveau -1...7, Toit).
- **Walls:** 92 total. I added 6 (IDs 258012-258017, all type "Ext. Voile BA 20"); 86 pre-existed.

What changed after step 3 vs. right after step 2:
- Right after step 2: all 6 walls = 2700 mm high.
- After step 3: exactly one wall changed – wall #3 (ElementId 258014, spanning x = 8000→12000 mm), 2700 mm → 2900 mm (+200 mm).
- The other five walls are unchanged at 2700 mm.

Final live state of the 6 walls: each 4000 mm long, laid end-to-end x=0→24000, on N0, base offset 0; heights = 2700, 2700, 2900, 2700, 2700, 2700.

Two honesty notes on the authoring (both corrected through Revit-direct tools, verified live):
1. `create_line_based_element` ignored my `baseLevel: N0` and placed all 6 walls on level "Toit" (elev 23750 mm). I corrected the base constraint to N0 via `batch_set_parameters` (`WALL_BASE_CONSTRAINT`).
2. That reassignment left a stray base offset, so I then set `WALL_BASE_OFFSET = 0`. Re-query confirms z-base = 0, z-top = height on all 6.

One caveat on provenance: the live model carries no edit history, so the "right after step 2" baseline (all 6 = 2700 mm) is asserted from my own create call and the fact that only wall #3 was touched in step 3 – not reconstructed from the model, which cannot provide it.

Reponse C (KG v2)

Done. All work was performed for real in the open Revit model and verified against the KG v2 projection.

Result

Note on scope: the open model is the pre-populated `500\_objs` project, so the totals below include pre-existing elements. I've separated what I authored from the model totals.

Exact counts (live project state)
I created (this task)	Total in model	
Levels	2 – `N0` @ 0 mm, `N1` @ 3000 mm	**12**
Walls	6 – `wall\_135`...`wall\_140` (Revit IDs 257973-257978)	**90**

My 6 walls are all type "Mur de base : Int. Plâtre 100 mm" (typeId 29324), 4000 mm long, on level N0, laid end-to-end along X from x=0 to x=24000.

What changed after step 3 (vs. state right after step 2)
Exactly **one** substantive change:

- **Wall #3** = `wall\_137` (Revit ID **257975**, x=8000→12000 mm): height **2700 mm → 2900 mm** (KG diff turn 60: `height` 2.7 → 2.9 m).

The other five walls are unchanged at 2700 mm. The action log shows two incidental no-op entries on the same turn for the neighbours `wall\_136` and `wall\_138` – these are Revit auto-rejoin touches where `before == after` (geometry only, no height field), not real edits.

Verification chain: `kg_v2_get_by_revit_id(257975)` confirms `height: 2.9`; `kg_v2_diff_since` isolates the single height delta; the full Wall query confirms only wall_137 deviates from 2.7 m among my six.

S4 – Neutres (§8.4-iv)

Reponse A (Revit-direct)

S4 – Atomic batch with intentional failure: PASS

****Did the batch fail?**** Yes, as expected. The single `create\_level` call carrying all 5 levels was rejected wholesale with a validation error – `S4\_BAD` (array index 3) was flagged for missing the required `elevation` (`"received": "undefined"`). Because the batch is validated as one unit before any element reaches Revit, the rejection was all-or-nothing: not even the 4 valid levels (S4_L1/L2/L3/L5) were created.

****How many levels exist now?**** ****12**** – identical to the baseline. The live model re-query shows the same 12 pre-existing levels (Niveau -1/0/1/2/3/4/5/6/7, Toit, N0, N1). ****No `S4\_` level leaked.****

Atomicity holds: invalid element → entire batch rolled back, model unchanged.

Reponse C (KG v2)

S4 result – atomicity holds.

- ****Did the batch fail?**** Yes, as expected. The single `create\_level` call carrying all 5 levels was rejected wholesale with `Input validation error ... data[3].elevation: Required` – the invalid `S4\_BAD` (no elevation) failed schema validation, so the call never reached Revit. I did ****not**** invent an elevation to fix it.

- ****Level count now: 12**** – identical to baseline. No `S4\_` leaked (no S4_L1/L2/L3/L5 persisted either). The 12 nodes are exactly the pre-batch set (level_001-level_012, unchanged turns/revit_ids).

Atomicity confirmed: one invalid element → none of the 5 persisted.

Q – Neutres (critere §8.4-iv)

Reponse A (Revit-direct)

All data now comes from live-model queries. Before the answers, two honest notes on what the Revit-direct tools could and couldn't do:

- ****Names****: Revit walls/windows have no user "name" instance parameter exposed by ``create_*`, so the labels `*Q_WALL_1..6*` and per-window names can't be stored in the model. I map each element to its label by ****geometry + ElementId**** (the only stable live identifiers).
- ****Level****: ``create_line_based_element`` binds a wall to a level `*by elevation*`, and an existing level ****N0**** already sat at elevation 0, so the walls first landed on N0. I corrected this with ``batch_set_parameters`` (`WALL_BASE_CONSTRAINT → Q_N0`); the live query now confirms all six on `Q_N0`.

****Q1 – Walls on Q_N0.**** Querying all walls (``ai_element_filter``, `OST_Walls`) and filtering by ``Level.Name``, exactly ****6**** walls report level `Q_N0` (Id 258029): ElementIds 258041, 258043, 258045, 258047, 258049, 258051. No other wall in the project is on `Q_N0`.

****Q2 – Average wall height of the Q_WALL_* walls.**** Each of the six walls reports the ``height`` (height) parameter = ****2700 mm**** (258041→2700, 258043→2700, 258045→2700, 258047→2700, 258049→2700, 258051→2700). Average = (2700×6)/6 = ****2700 mm****.

****Q3 – Windows and their host wall.**** Three windows exist on the `Q_N0` row (type 175114, all at LocationY 0). By their queried LocationX (in feet) mapped onto the wall spans: window ****258054**** (x≈2.0 m / 2000 mm) → ****Q_WALL_1**** (258041, span 0–4000); window ****258055**** (x≈10.0 m / 10000 mm) → ****Q_WALL_3**** (258045, span 8000–12000); window ****258056**** (x≈18.0 m / 18000 mm) → ****Q_WALL_5**** (258049, span 16000–20000). (Each x falls inside exactly one wall, so the host is unambiguous.)

****Q4 – Q_WALL_* walls with no window.**** The windowed walls are 1, 3, 5; therefore the walls with ****no window**** are ****Q_WALL_2** (258043), ****Q_WALL_4** (258047), ****Q_WALL_6** (258051)**.

****Q5 – Total wall length on Q_N0.**** From each wall's queried bounding box, every wall spans exactly 4000 mm in X (0→4000, 4000→8000, 8000→12000, 12000→16000, 16000→20000, 20000→24000). Sum = 6 × 4000 = ****24000 mm**** (24.0 m).

Summary: ****Q1=6, Q2=2700 mm, Q3=(258054→Q_WALL_1, 258055→Q_WALL_3, 258056→Q_WALL_5), Q4=(Q_WALL_2, Q_WALL_4, Q_WALL_6), Q5=24000 mm****

Reponse C (KG v2)

Seed complete. The window height is type-driven on the only available family (fixed 0.95 m tall), so head stays at 1.85 m – not achievable as 2.1 m without a different family. Sill is correct at 0.9 m. This doesn't affect any of Q1-Q5. Answers below, all read back from the KG/Revit tools (not from creation memory):

****Q1.**** Traversing ``at_level`` into level ``Q_N0`` (level_013, revit_257981) returns exactly ****6 walls**** (wall_141–wall_146), each with ``level_ref: revit_257981``. So ****6 walls**** exist on `Q_N0`.

****Q2.**** Every `Q_WALL_*` node reports ``height: 2.7`` m. Average = (2.7×6)/6 = 2.7 m = ****2700 mm****. All walls are uniform height, so the average equals each individual height.

****Q3.**** Three Window nodes exist, each via its ``host_wall_ref``: ****window_353 → Q_WALL_1**** (host revit_257993), ****window_354 → Q_WALL_3**** (host revit_257997), ****window_355 → Q_WALL_5**** (host revit_258001). Each sits at the hosting wall's midpoint (x = 2 m, 10 m, 18 m respectively).

****Q4.**** Cross-referencing the three host walls (141/143/145 = WALL_1/3/5) against the full set of 6, the walls with ****no window**** are ****Q_WALL_2, Q_WALL_4, and Q_WALL_6**** (wall_142, wall_144, wall_146 – none appear as a ``host_wall_ref``).

****Q5.**** Each of the 6 walls reports ``length: 4`` m (4000 mm); summing across all `Q_WALL_*` gives 6 × 4000 = ****24000 mm****.

****Summary:**** ``(6, 2700, [window_353→Q_WALL_1, window_354→Q_WALL_3, window_355→Q_WALL_5], [Q_WALL_2, Q_WALL_4, Q_WALL_6], 24000)``

M – Neutres (critere §8.4-iv)

Reponse A (Revit-direct)

Result

****Step 1 – Seed:**** Created level `M\_N0` (elevation 0) and 6 walls of type **Mur de base – Ext. Voile BA 20**, 200 mm thick, 2700 mm high, end-to-end along X. Element IDs: M_WALL_1=258078, M_WALL_2=258081, M_WALL_3=258084, M_WALL_4=258087, M_WALL_5=258090, M_WALL_6=258093.

****Step 2 – 10 height edits:**** Applied in order via `batch\_set\_parameters` (one persisted call each); every call returned `succeeded:1, failed:0, rolled\_back:false`.

****Step 3 – Final heights (live model):****

Wall	Element ID	Edits applied (in order)		Final height
M_WALL_1	258078	2701 → 2710	**2710 mm**	
M_WALL_2	258081	2702 → 2720	**2720 mm**	
M_WALL_3	258084	2703 → 2730	**2730 mm**	
M_WALL_4	258087	2704 → 2740	**2740 mm**	
M_WALL_5	258090	2705	**2705 mm**	
M_WALL_6	258093	2706	**2706 mm**	

****Summary:** 10 height edits were applied and committed across the 6 walls (M_WALL_1-4 each edited twice, M_WALL_5-6 once).

Verification note (honest scope): the live query tool returned the height directly for M_WALL_1 = 2710 mm. The other five could not be re-read in the same call because `ai\_element\_filter`'s bounding-box filter is non-functional in this build (returns a fixed view-clipped subset regardless of the box) and its unfiltered list caps at 50 elements ordered below my element IDs; `get\_selected\_elements` returns no parameter values. The five values above are therefore reported from the committed (non-rolled-back) transactions plus the direct read of M_WALL_1. Separately, the create tool hosted the walls on a pre-existing level named "N0" that also sits at elevation 0, rather than the new "M_N0" – `create\_line\_based\_element` resolves by elevation value, which is ambiguous when multiple coincident levels exist at 0.

Reponse C (KG v2)

All 10 edits are persisted and verified. Final heights from the KG v2 projection:

Wall	Revit ID	Final height
M_WALL_1	258028	2710 mm
M_WALL_2	258031	2720 mm
M_WALL_3	258034	2730 mm
M_WALL_4	258037	2740 mm
M_WALL_5	258040	2705 mm
M_WALL_6	258043	2706 mm

These match the expected end state: walls 1-4 reflect their second-round edits (2710/2720/2730/2740), walls 5-6 retain their single edit (2705/2706).

****Summary:** 10 height edits applied (6 walls created on a level at elevation 0). **** Note:** `create\_line\_based\_element` binds walls by elevation only, so Revit hosted them on the pre-existing level at elevation 0 (`N0`, revit_id 257950) rather than the freshly created `M\_N0` – the heights are unaffected.

Impact-XL – Graphe-shaped (avantage structurel)

Reponse A (Revit-direct)

Walked the dependency chain via the live model. Here is the Niveau 0 blast radius.

Niveau 0 – structural blast radius

Hop	Element type	Count
1	Walls on Niveau 0	10
2	Windows hosted on those walls	44
	Total blast radius	54

- Walls on Niveau 0: 10 (level Id 608) – 4 exterior `G n rique - Ext. 200 mm` + 6 `cloison` partitions. (Whole model has 84 walls; only these 10 sit on Niveau 0.)
- Windows transitively hosted on those walls: 44 – all `0.60 m x 0.95 m` openings whose level association is Niveau 0, sitting at sill z=820 on the Niveau-0 walls.
- Total: 10 + 44 = 54 elements orphaned/destroyed.
- By hop distance from the level: hop 1 = 10 (walls), hop 2 = 44 (windows). No deeper hops – windows host nothing further.

Method / honest caveat: I queried `OST\_Walls` and `OST\_Windows` directly and grouped by their associated `Level`. `ai\_element\_filter` does **not** expose a window's actual host element, so hop 2 is computed from each window's level association rather than by reading the literal `host wall Id`. In this model that is faithful – every window inherits its host wall's level, and the 44 windows' XY positions line up exactly with 4 of the 10 Niveau-0 walls (the two exterior 200 mm walls at x=4985 / x=-9014 and the two interior cloisons at x=885 / x=-4914). If you need the literal per-window host-wall binding confirmed (not just level inheritance), that requires reading `FamilyInstance.Host`, which these query tools don't return – say the word and I'll note it can't be had through `ai\_element\_filter`/`get\_` alone.

Nothing was modified.

Reponse C (KG v2)

Niveau 0 – structural blast radius

Walked `level\_001` ("Niveau 0") → walls (`at\_level`, incoming) → windows (`hosts`, outgoing) entirely in the KG v2 projection. The fixed-path traversal confirms the transitive hosting chain.

#	Quantity	Count
1	Walls on Niveau 0	10
2	Windows transitively hosted on those walls	44
3	Total blast radius (walls + windows)	54

Breakdown by hop distance from the level:

Hop	Edge followed	Node type	Count
1	level ← `at_level` ← wall	Wall	10
2	wall → `hosts` → window	Window	44
	Total		54

Deleting "Niveau 0" would destroy its 10 hosted walls (`wall\_001`-`wall\_010`), and orphan/destroy the 44 windows those walls host (`window\_001`-`window\_044`) – 54 dependent elements in all.

Note: a single variable-depth reachability sweep (`edge\_types=[at\_level, hosts]`, `direction=in`) returned the same 54-node set, but reported every node at `depth 1` because each window also carries its own direct `at\_level` edge to the level. The explicit two-step path above is what cleanly separates the hops and proves the 44 windows are reached *through* the walls. Nothing was modified.

TypeImpact-XL – Graphe-shaped (avantage structurel)

Reponse A (Revit-direct)

****TypeImpact-XL – window ripple ranking**** (live model `500\_objs`: 84 walls, 352 windows, all hosted on walls – `noHost=0`).

Chain used: `WallType → Wall.GetTypeId()` for instance counts, `Window.Host → Wall → GetTypeId()` for the transitive window ripple. Two WallTypes are in use:

Rank	WallType (id)	Name	Wall instances	Windows hosted (ripple)
1	29331	Générique - Ext. 200 mm	36	**240**
2	257865	cloison	48	**112**

(240 + 112 = 352 = all windows, fully accounted for.)

****Largest window ripple: WallType `Générique - Ext. 200 mm` (id 29331) – 240 windows.**** Note it ranks first on ripple despite having **fewer** wall instances than `cloison` (36 vs 48), so window density per wall, not wall count, drives the impact.

Reponse C (KG v2)

All data gathered – 352 windows total, all resolved to a host wall type (240 + 112 = 352). Wall instance counts per type (from the 84-wall join: 9 blocks of 4 for `walltype\_013`, 8 blocks of 6 for `walltype\_021`, = 36 + 48 = 84 ☑).

TypeImpact-XL – Window ripple by WallType

Chain used: ****WallType → walls**** (`is\_type`) ****→ windows**** (`hosts`), computed entirely from the KG v2 projection (2 join queries + offline counting; no per-wall re-query).

Rank	WallType id	Name	Wall instances	Windows (ripple)
1	`walltype_013`	Générique - Ext. 200 mm	36	**240**
2	`walltype_021`	cloison	48	**112**

Only these two WallTypes are actually instantiated (84 walls total). Note the inversion: `walltype\_021` has more wall instances (48 vs 36) but a smaller window ripple – windows concentrate on the exterior type.

****Largest window ripple: `walltype\_013` ("Générique - Ext. 200 mm") – 240 windows.****

Provenance – Graphe-shaped (avantage structurel)

Reponse A (Revit-direct)

I built the chain through the live model. Now Step 2/Step 3.

****Step 2 – attempting backward reconstruction from the final living wall (257957):****

Starting only from wall ****257957**** and querying the live Revit model, I look for any attribute, parameter, or relationship that points to a predecessor. There is none. A Revit `Wall` carries geometry, type, level, and instance parameters – it has ****no field referencing a prior deleted element****, and the three predecessors (257954, 257951, 255040) no longer exist in the model (each `delete\_element` returned `deleted: true`; they are tombstoned and unqueryable).

****Step 3 – explicit limitation statement:****

My tools ****cannot**** traverse from a live element back through deleted predecessors. The "supersession chain" this task asks for is a ***semantic*** relation (`replaces` + reason), and the live Revit model has no place to store or recover it:

- Deleted elements leave no tombstone the live API can read – once `delete\_element` runs, the id is simply gone.
- A newly created wall holds no provenance link to the wall it conceptually replaced; same position/level is not a recoverable relation.
- The only layer that could persist `replaces` edges and reasons across deletion is a project-memory/KG layer – which this profile forbids (`store\_\*/kg\_\*` disallowed), and I was instructed not to reconstruct from conversation memory.

So I will not fabricate predecessor ids, alive/dead status, or reasons. The only fact the live model can confirm is the single surviving wall.

Recoverable from the live model alone: ****1 node**** (the living wall 257957). The three-deep predecessor history is ****not recoverable**** by Revit-direct tools.

supersession depth: 1 recovered (1 alive, 0 deleted)

Reponse C (KG v2)

All three reasons are now confirmed directly from the action log (turns 54, 56, 58), independent of Step 1 memory. The provenance traversal succeeded across deletions, so Step 3's "cannot traverse" clause does not apply.

Step 2 – Supersession history, reconstructed from the live wall

Starting node: ****wall_137**** (revit_id 257959) – ***alive***, type ***Générique - Ext. 200 mm***. Walking `replaced\_by` edges in the `in` direction with `include\_soft\_deleted=true` recovered the full chain transitively:

Order back from live	Node	Revit id	State	Replacement reason (recorded)
--- --- --- --- ---	start	**wall_137**	257959	☑ alive – (current living wall)
depth 1	**wall_136**	(257955)	tombstoned	"Replaced Int. Brique 110 mm with Générique - Ext. 200 mm type (round 3)"
depth 2	**wall_135**	(257951)	tombstoned	"Replaced Int. Plâtre 70 mm with Int. Brique 110 mm type (round 2)"
depth 3	**wall_001**	(255040)	tombstoned	"Replaced original wall with Int. Plâtre 70 mm type (round 1)"

Read as a forward chain: ****wall_001**** (original, deleted) → ****wall_135**** (deleted) → ****wall_136**** (deleted) → ****wall_137**** (alive).

The `replaced\_by` annotation edges persisted across each deletion: the traversal reached every tombstoned predecessor purely from edge structure (not from prior knowledge), and each edge's payload – including the one-line reason – was recoverable from the action log. Provenance survives deletion by design here, so no fabrication was needed.

supersession depth: 4 recovered (1 alive, 3 deleted)